SST - BASED SMART HELMET FOR CYCLISTS AND MOTOR CYCLISTS

25-26J-294

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Overall Title and Content

HEALTH MONITORING SYSTEM

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HEALTH MONITORING SYSTEM

Proven Gap

- Creative solution: Multi-sensor fusion helmet with PLUSE SENSOR + TEMPERATUE SENSOR + EEG analysis for **proactive health intervention**.
- Predict heart attack and stress risks in real-time.

Creative Solution

A multi-sensor fusion–based health monitoring component that integrates heart rate, body temperature, and EEG signals, processes them on an ESP32 with edge-based machine learning, and delivers real-time alerts, hands-free emergency commands, and personalized health coaching during rides.

Knowledge Gap

- Gap: Existing systems lack integrated real-time detection of physiological + cognitive states during rides.
- Problem: Riders remain unprotected from sudden cardiovascular and stress-related incidents.

Shortcomings of Existing Systems	Proposed System
Focus only on accidents (GPS/GSM) - no health monitoring.	Provides **multi-sensor real-time analysis** on ESP32.
Single-parameter monitoring – no predictive insights.	Enables **hands-free emergency response**.
No post-ride health analytics	Generates **post-ride health trend reports** for early intervention.

HEALTH MONITORING SYSTEM

Domains of Knowledge

- Embedded Systems & IoT
- Biomedical Signal Processing (HRV, EEG ratios)
- Machine Learning (TinyML, Edge ML)
- Mobile App Development (Flutter, Firebase)

Latest Technologies Used

- ESP32 microcontroller with TinyML models
- MAX30102 pulse sensor, LM35 temperature sensor, EEG module
- BLE for real-time data transfer
- Firebase + Flutter for visualization

HEALTH MONITORING SYSTEM

Evaluation & Validation

Accuracy Metrics : Pulse sensor, Temperature sensor and EEG-based emotional state detection validated with **Precision, Recall, and F1-score**.

Pilot Testing : Compare helmet readings with **clinical devices** (ECG, digital thermometer) and rider self-reports for real-world reliability.

Expected Accuracy : 80–85% correct detection of irregular heartbeat, heat stress, and cognitive overload.

Data Availability & Ethics

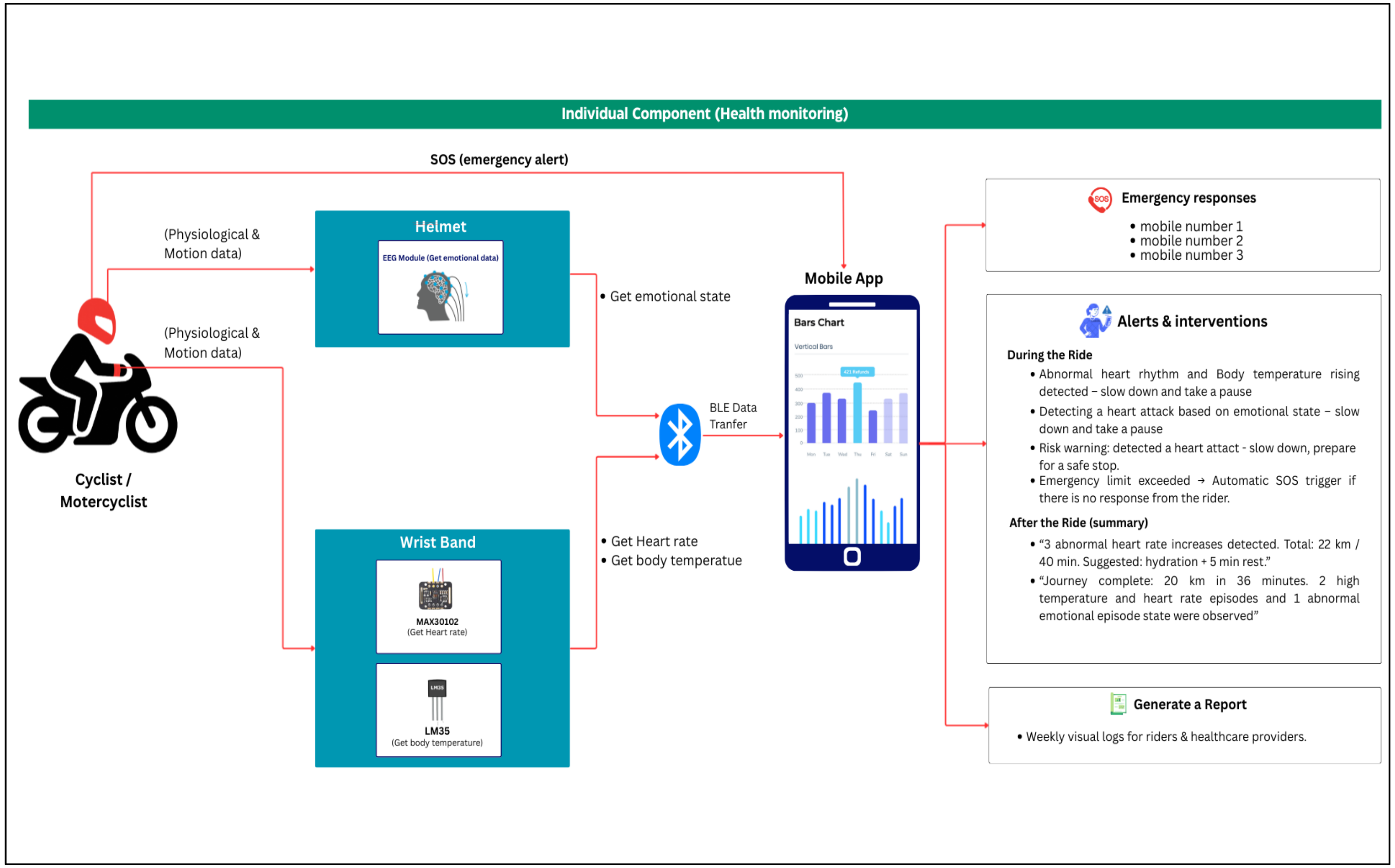
Datasets : Use **DEAP, WESAD** (stress/EEG) + **locally collected HRV & temperature logs** from rider trials.

Sensor Data : MAX30102 (heart rate/HRV), LM35 (temperature), EEG modules embedded in helmet.

Ethical Compliance : Rider health data **encrypted, anonymized, and used only with informed consent**.

Privacy Compliance: On-device (edge) processing ensures sensitive data is **not sent to cloud**, reducing exposure risks.

HEALTH MONITORING SYSTEM



Real-World Scenario

- Rider shows abnormal HRV + high EEG stress signals → system issues **voice alert**.
- Rider can say **"Send Help"** → auto-call emergency contact.
- After journey → mobile app shows weekly trend graphs and risk alerts.

Emotion and Stress Monitoring with Real-Time Feedback

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Emotion and Stress Monitoring with Real-Time Feedback

Proven Gap

- Most existing smart helmets focus on physical safety (accident detection, crash alerts), but do not address the rider's **mental stress and emotional state** in real-time.
- **Creative solution**: An integrated **EEG-based stress detection component** that captures brainwave activity, analyzes stress using personalized machine learning, and provides **non-distracting interventions** during rides.

Knowledge Gap

- **Problem:** Riders remain unprotected from sudden stress-induced risks, as current solutions do not provide safe, real-time detection or rider-friendly interventions during journeys.
- **Gap:** Conventional helmet systems lack integrated real-time EEG-based stress monitoring due to motion artifacts, comfort limitations, and the absence of wearable-friendly designs.

Shortcomings of Existing Systems & How Proposed Solution Addresses Them

Shortcomings of Existing Systems	Proposed System
Limited to lab settings → fail in motion-heavy environments	Motion artifact removal and preprocessing ensure cleaner EEG during rides.
Artifact contamination & poor signal quality → unreliable in daily use	Low-latency BLE transmission for continuous data flow to a mobile app.
Lack of BLE-based streaming → high latency or wired connections	Personalized ML stress model adapts to individual brainwave patterns.
No personalized stress models → poor generalization across users	Non-distracting interventions (subtle vibration, soft audio, friendly notifications with time & distance context).

Emotion and Stress Monitoring with Real-Time Feedback

Domains of Knowledge

- **Biomedical Signal Processing** – EEG filtering and artifact removal.
- **Machine Learning** – Stress classification with CNN/LSTM models.
- **Wireless Communication** – ESP32 + BLE 5.0 for real-time data transfer
- **HCI & Visualization** – Rider-safe alerts and mobile dashboard (Flutter + Firebase).

Latest Technologies Used

- **ESP32 MCU with BLE 5.0** for real-time EEG data transmission
- **EG Module (dry electrodes)** for continuous brainwave capture (Delta, Theta, Alpha, Beta, Gamma)
- **Artifact Removal Pipeline** using Band-pass filters + ICA + Adaptive Filtering
- **TensorFlow Lite / ONNX Runtime** for lightweight inference on mobile
- **Flutter + Firebase** for cross-platform app and visualization dashboard

Emotion and Stress Monitoring with Real-Time Feedback

Evaluation & Validation

Model Validation: Accuracy, precision, recall, and F1-score for calm vs stressed classification.

Latency Check: BLE transmission delay <100 ms to ensure real-time response.

Usability Testing : Rider surveys on comfort, distraction level, and effectiveness of interventions.

Expected Accuracy : ≥90% with personalized calibration and artifact removal.

Data Availability & Ethics

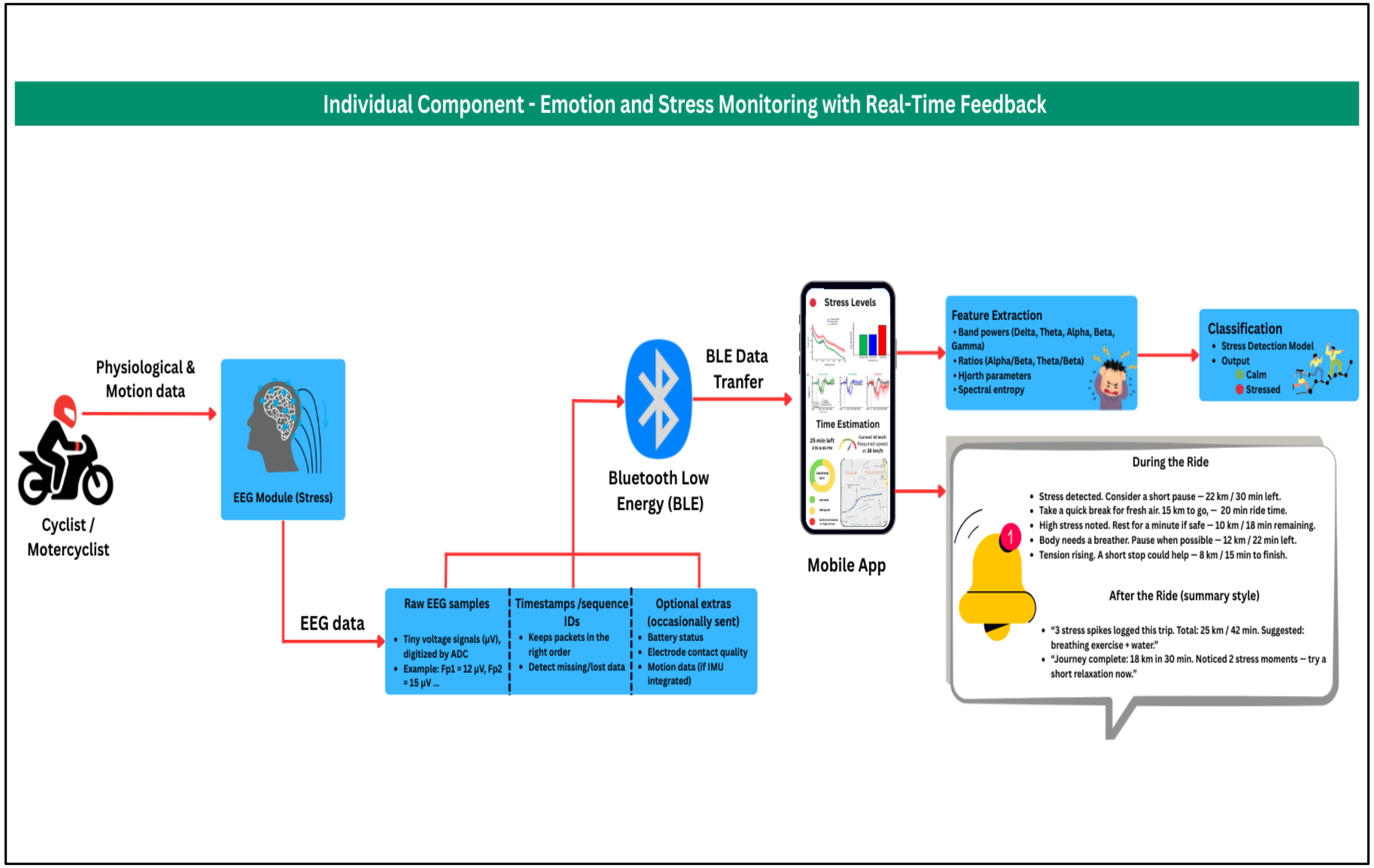
Datasets: DEAP, DREAMER, SEED datasets with pilot helmet EEG rides for stress model training.

Sensor Data: EEG dry electrodes plus IMU, battery, and electrode quality monitoring during rides.

Ethical Compliance: Rider EEG data encrypted, anonymized, and collected only with proper informed consent.

Privacy Compliance: On-device inference ensures raw EEG stays local, only stress summaries securely stored.

Emotion and Stress Monitoring with Real-Time Feedback



Real-World Scenario

Before Ride: Rider wears the smart helmet; mobile app pairs automatically via BLE.

During Ride:
- EEG data streams continuously.
- App detects stress episodes in real time.
- Rider receives safe alerts (gentle vibration, friendly notification with time & distance left).

After Ride:
- Summary report generated (stress spikes, ride time, relaxation suggestions).
- Data logged for long-term tracking and personalization.

Post Journey Risk Assessment And Report Generation

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Post Journey Risk Assessment And Report Generation

Proven Gap :

Existing smart helmets mostly stop at real-time alerts (fall detection, heart rate warnings). They fail to provide post-journey insights combining riding behavior, stress, and environmental risks.

Creative Solution :

AI-powered risk assessment engine that synchronizes IMU + EEG + environmental data, generating personalized post-ride reports highlighting unsafe zones, stress triggers, and long-term safety trends.

Knowledge Gap :

- Lack of post-ride reports linking IMU events (sudden turns, over speeding) with EEG stress peaks.
- Limited use of contextual data (weather/traffic APIs) for risk interpretation.

Shortcomings of Existing Systems

- Current helmets end with real-time warnings only no follow-up analysis.
- No connection between stress data and riding behavior.
- Reports (if any) are raw sensor logs, not user-friendly insights.

Proposed Solution

- Fusion of EEG + IMU + Environmental APIs for comprehensive post-ride analysis.
- Risk visualization dashboards in a mobile app (heatmaps, stress peaks, safety scores).
- Personalized recommendations to improve rider safety over time.

Domains of Knowledge

- Data Science & Machine Learning (multimodal fusion, anomaly detection)
- Biomedical Signal Processing (EEG stress analysis, HRV correlation)
- Embedded & Mobile Computing (ESP32 edge processing, Firebase cloud sync)
- Data Visualization & UX (risk dashboards, report generation)

Latest Technologies Used

- **ML Models :** Random Forest, LSTM, Anomaly Detection (Isolation Forest/DBSCAN)
- **Visualization :** Flutter + Firebase + Chart.js/Plotly
- **APIs :** OpenWeatherMap, Google Traffic for context-aware risk mapping
- **Cloud :** Firebase Firestore for data sync, TensorFlow Lite for on-device ML

Evaluation & Ethical Commitment

Evaluation & Validation

Accuracy Metrics : We measure Precision and Recall for anomaly detection, and F1-score for comprehensive event classification

Pilot Testing : Detected risk zones are validated against ground-truth GPS data and direct rider feedback for real-world relevance.

Expected Accuracy : We anticipate 80-85% correct identification of stress-risk events, ensuring high reliability for riders.

Data Availability & Ethics

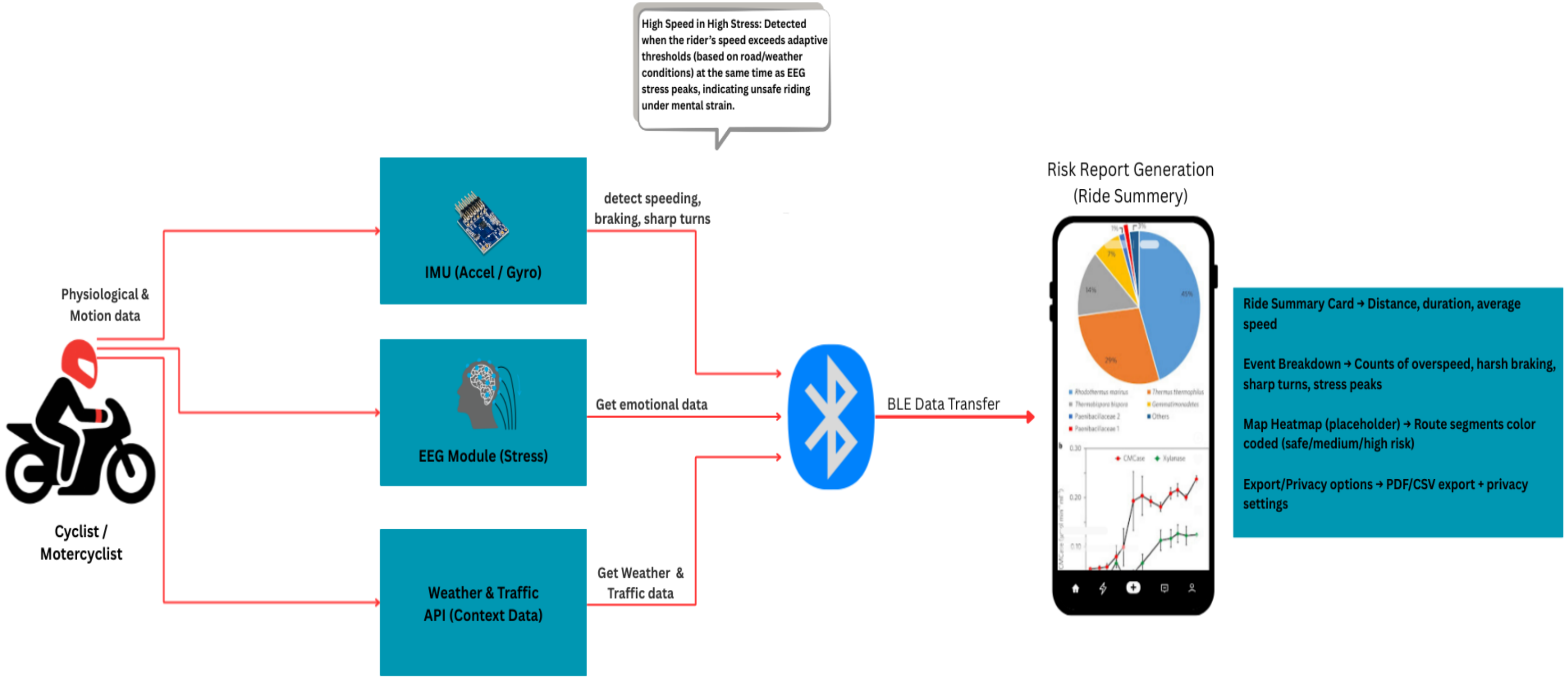
EEG & IMU Datasets : Utilizing a blend of public datasets and locally collected riding trial data for robust model training

External Data : Contextual data is enriched using reliable Weather and Traffic APIs, providing crucial environmental factors.

Ethical Compliance : Our policy mandates anonymized rider data, clear informed consent, and adherence to stringent GDPR-style privacy regulations.

Project Completion

Individual Component - Post Journey Risk Assessment and Report Generation



Real-World Scenario

- A rider completes a 30 km trip.
- One sharp turn executed under heavy traffic conditions
- One overspeed incident during a period of heavy rain.

Danger Zone Detection & Voice-Guided Riding Assistant

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Danger Zone Detection

Proven Gap :

- Most rider apps warn only after entering a risky stretch; our module predicts upcoming danger zones and speaks proactive guidance (Sinhala/English) with distance-aware prompts.

Creative Solution :

- Combine **IMU + EEG + APIs** to generate personalized, context-aware safety reports.
- Highlight **stress peaks, risky turns, speed in unsafe zones**.
- Store historical data for **weekly/monthly insights**.

Knowledge Gap :

- Lack of integration between riding behavior (IMU) and emotional/health states (EEG).
- Existing reports don't link external conditions (traffic, weather) with rider performance.

Shortcomings of Existing Systems

- Existing maps lack rider-specific, sensor-aware risk (IMU/EEG) context
- Slow, linear geofence checks.
- Text-only warnings

Proposed Solution

- We blend GPS, IMU, and (optionally) EEG for personalized risk zones.
- We use R-tree/KD-tree spatial indexing for millisecond-level lookups.
- **TTS voice** alerts in Sinhala/English; safer alternate path suggestions pre-ride.

Domains of Specialization

- IoT & Sensor Fusion – IMU + EEG data combination
- Data Analytics – behavior + stress risk profiling
- Mobile Development – visualization in reports
- Geospatial APIs – weather + traffic context
- BLE sync to helmet MCU (ESP32)
- Geospatial indexing & route analysis (R-tree/KD-tree, map SDKs)

Technologies Utilized

- BLE for live updates, ESP32 edge processing
- Firebase (cloud storage + sync)
- Mapbox/Google Maps SDK (journey maps + risk visualization)
- TensorFlow Lite (risk pattern analysis)
- Text-to-Speech (Google TTS/Amazon Polly)

Evaluation & Ethical Commitment

Evaluation / Success Proof

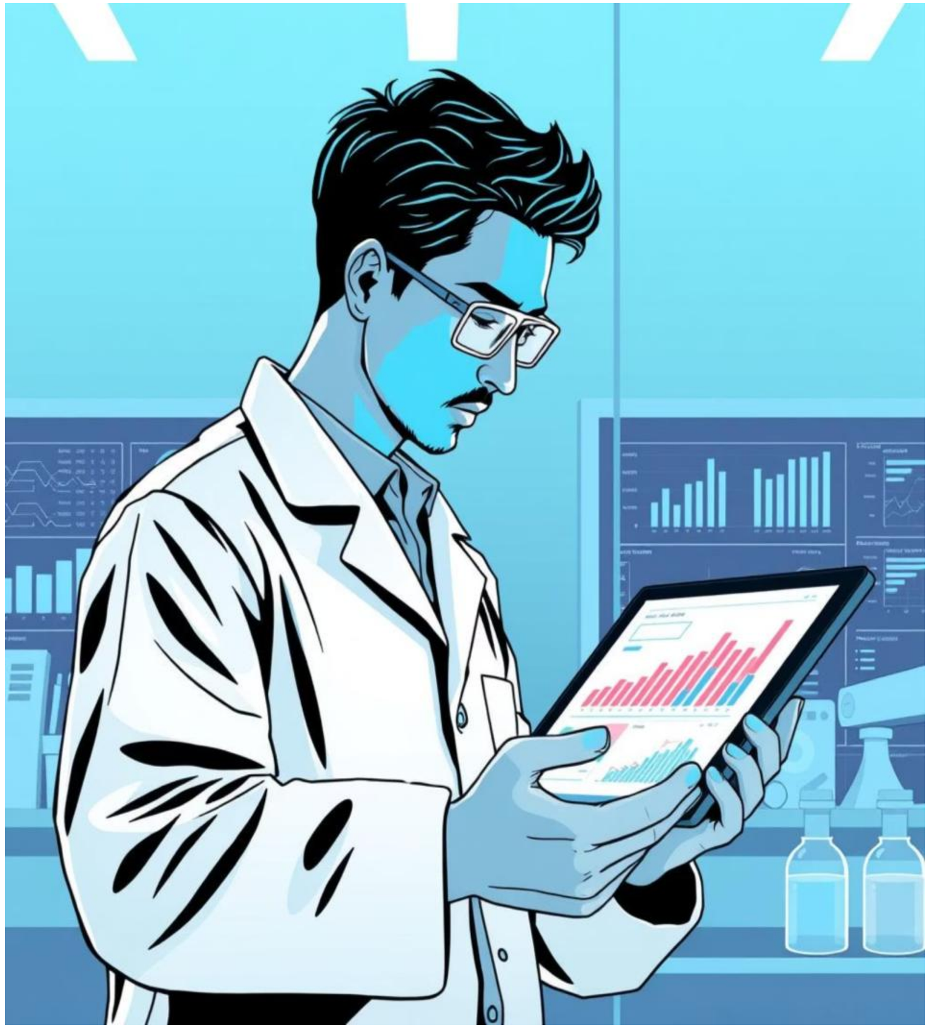
- Compare risky event detection vs. real ride data.
- Accuracy of IMU-based anomaly detection (>85%).
- Rider usability feedback (reports clarity: 4/5).

Data Availability

- Historical ride logs (GPS/IMU; optional EEG stress peaks)
- Traffic & weather APIs (Google Traffic, Open Weather).
- Public accident/Road-hazard datasets.
- Details under another riders.

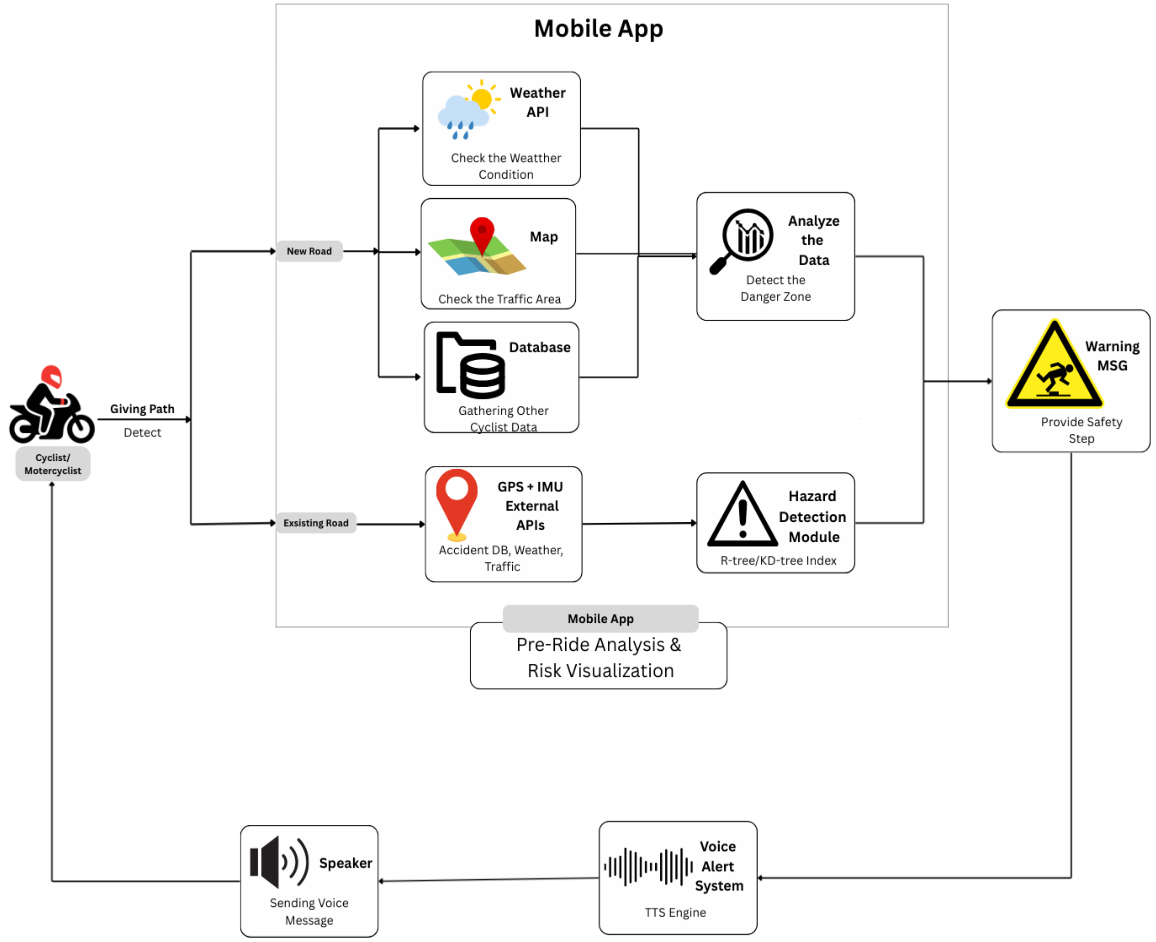
Ethical Clearance & Privacy

- **Informed Consent:** Mandatory rider consent for all data collection and processing.
- **Anonymized Data:** Strict anonymization of all stored ride histories to protect privacy.
- **Rider-Only Access:** Reports and personal data are restricted solely for the rider's use and insight.



Completion of the project

System Overview



Real-World Scenario

Consider a rider completing a challenging 20km urban ride. The system's report concisely flags critical safety events:

- ⚠ "High speed + high stress detected in rainy conditions."
- ⚠ "Multiple sharp turns identified within dense traffic zones."
- Rider adjusts future routes & learns safer riding habits.

SST - BASED SMART HELMET FOR CYCLISTS AND MOTOR CYCLISTS

Target Market & Customer Profile

- Primary Market: Urban motorcyclists & cyclists in Sri Lanka & South Asia.
- Early adopters: Delivery riders (PickMe, Uber Eats, courier fleets).
- Secondary: Commuters, safety-conscious riders, student riders.
- B2B Clients: Insurance companies, transport authorities, healthcare providers.

Value Proposition

- Transforms a passive helmet → AI-powered safety companion.
- Prevents accidents with real-time alerts and post-ride risk insights.
- Enables early stress/fatigue detection and long-term health awareness.
- Provides data-driven evidence for insurance claims & policy adjustments.

Ability of Commercialization

- Wearable IoT-based safety device with real-time monitoring + post-ride risk analytics.
- Scalable to delivery fleets, insurance companies, and smart mobility markets.
- Strong potential for B2C sales (commuters) and B2B partnerships (insurance, ride-hailing, logistics).

Investment / Cost & Recovery

- Labor + Material Cost: Approx. Rs. 15,000 – 18,000 per unit (Helmet + EEG + IMU + ESP32 + app).
- B2C pricing: Rs. 25,000 – 30,000 per helmet (retail margin included).
- B2B subscription: SaaS dashboard for fleets & insurers (Rs. 2,000–3,000/month).
- Break-even expected in 1.5 – 2 years with local + regional adoption.

Intellectual Property Rights (IPR)

- Patentable **Post-Journey Risk Analytics Engine (EEG + IMU fusion + API integration)**.
- Software copyrights (mobile app, visualization dashboards).
- Trademark opportunity for helmet brand under **Smart Helmet Initiative**.

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